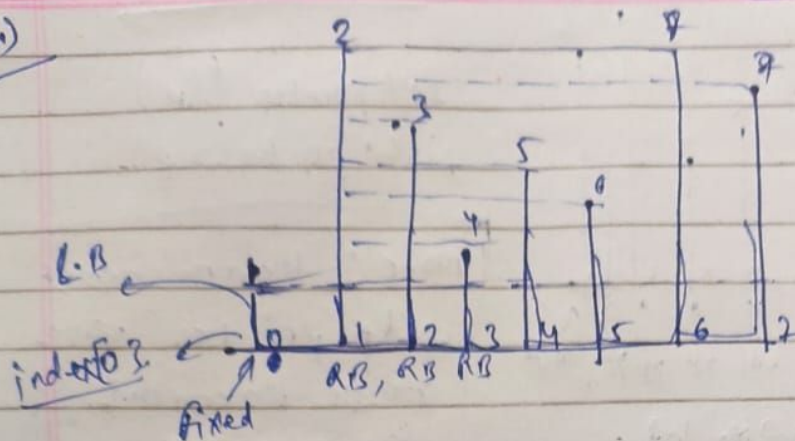


2

Max. water can store

brute force code

P100)



To store max water we need two lines L.B

X R.B (right boundary)
the smaller boundary decided to store max water.

now, fixed index [1].

L.B = 2.

R.B = 3, 4, 5, 6, 7, 8. (decide water store).

Fixed index [2].

L.B = 3

R.B = 4, 5, 6, 7, 8.

Fixed index [3].

L.B = 4

R.B = 5, 6, 7, 8

Fixed index [4].

L.B = 5

R.B = 6, 7, 8

Fixed index [5].

L.B = 6

R.B = 7, 8

loop exit.

Ish tarah hai

L.B X R.B ko check
karona hai or max.
water store nikalna
hai.

code:-

```
for (i = 0; i < n; i++) {
    for (j = i + 1; j < n; j++) {
        width = j - i;
        height = min(height[i], height[j]);
        area = width * height;
        maxWater = max(maxWater, area);
    }
}
```

return maxWater;

i = left boundary

j = Right boundary

* For Area we need width

$j - i = \text{width}$.

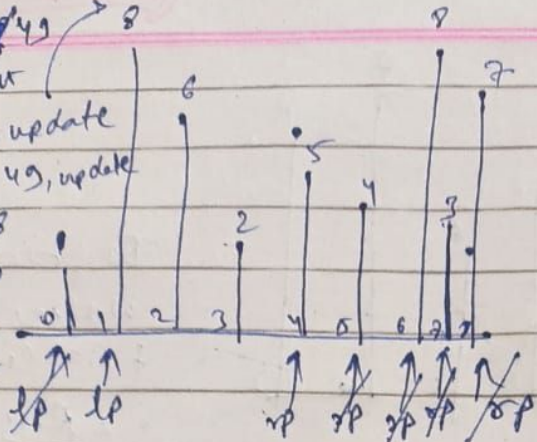
or $R.B - L.B = \text{width}$

Case III
 13 → usse chota hai
 no change only
 update.

30 chota hai
 wo update nahi
 hoga

optimal approach
or 2 pointer,

max water = $8 \times 1 = 8$, update
 case II = $(7-1) \times 7 = 49$, update
 case III = $(7-1) \times 3 = 18$
 case IV = $(6-1) \times 5 = 25$
 case V = $(5-1) \times 4 = 16$
 case VI = $(4-1) \times 5 = 15$



left pointer (lp)
 right pointer (rp)

Note:- karam height
 wala pointer water
 storage ko control
 karega or wo
 pointer ko
 update karne se
 bada pointer update
 nhi hoga.

Container = width $\times$ height

right $\times$ left

$(8-1)$
 or $(j-i)$

$\min(\text{ht}[l], \text{ht}[r])$

lp min $\rightarrow$ lp ++
 rp min $\rightarrow$ rp --

initialize.

```

lp = 0, rp = n-1, maxwater = 0
loop, while (lp < rp) {
    w = rp - lp;
    ht = min(ht[lp], ht[rp]);
    current = w * ht;
    maxwater = max(maxwater, current);

    if (ht[lp] < ht[rp]) {
        lp++;
    } else {
        rp--;
    }
}
  
```